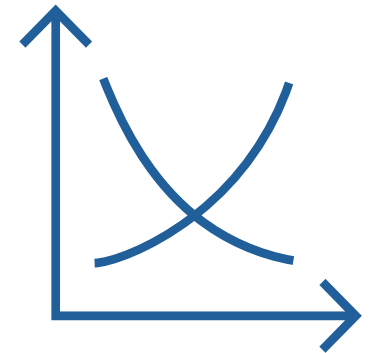
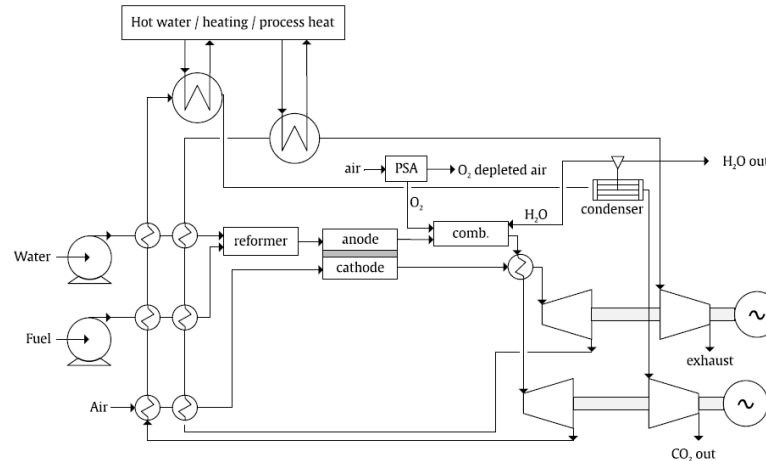
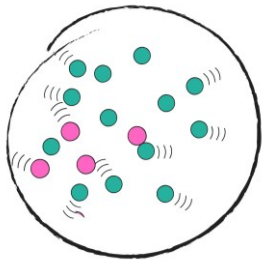
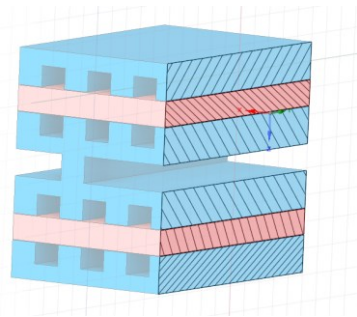
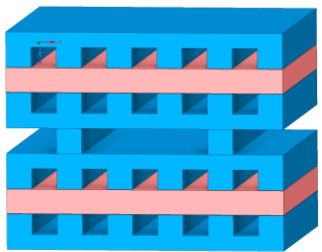


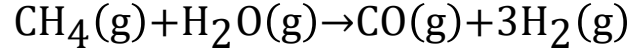
Hybrid SOFC-IBC with CO₂ Separation

Component Analysis

Shreekrishna Meher



1. Steam Methane Reforming



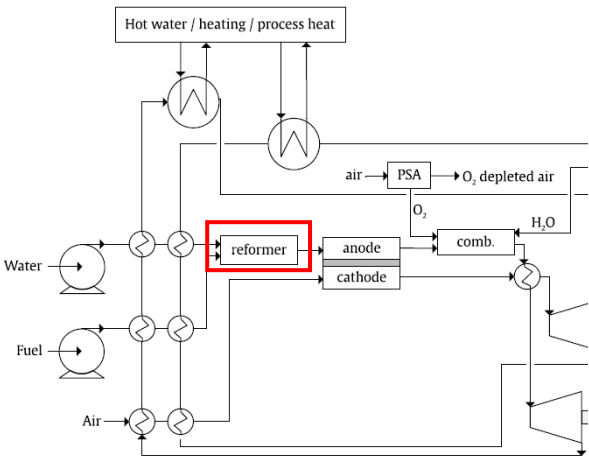
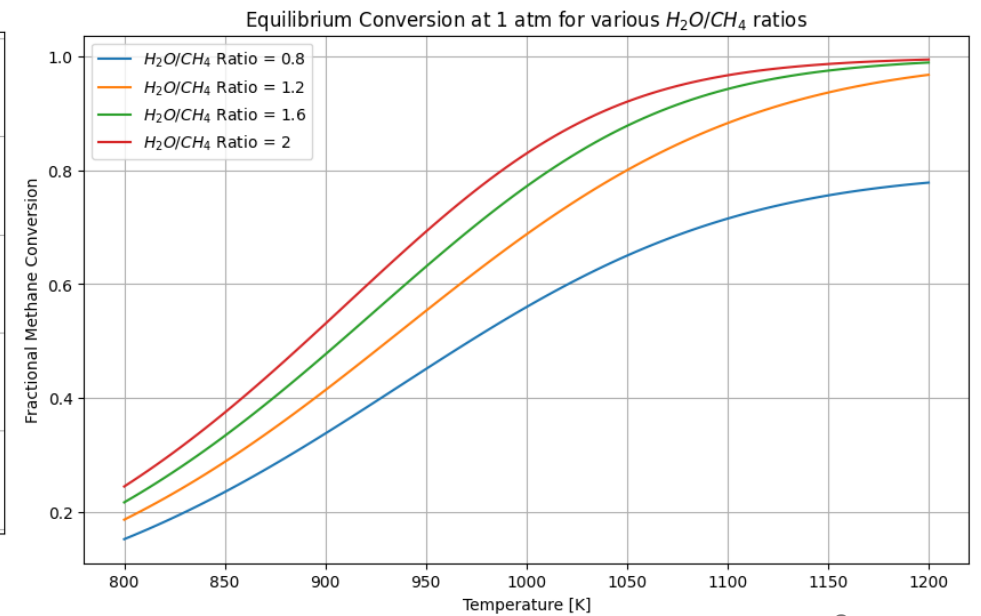
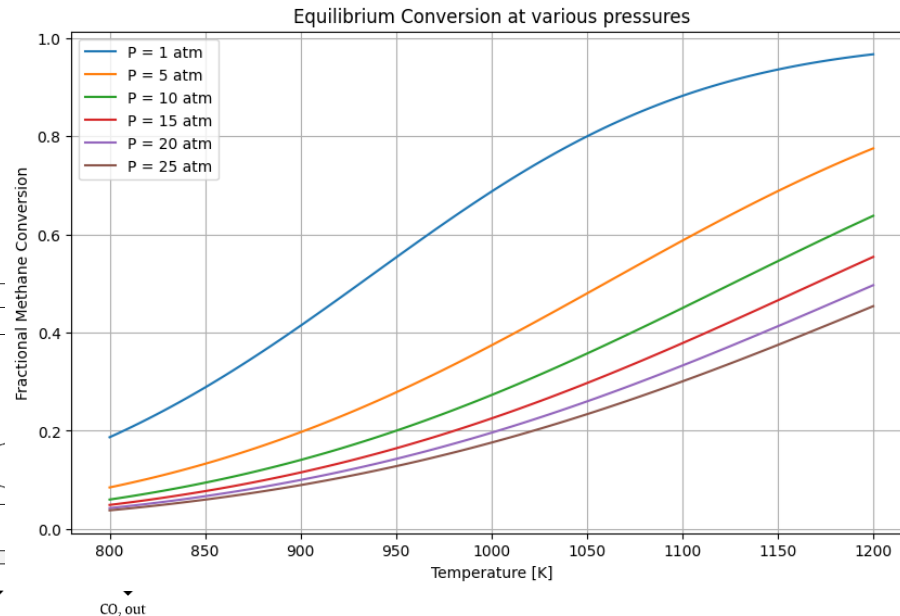
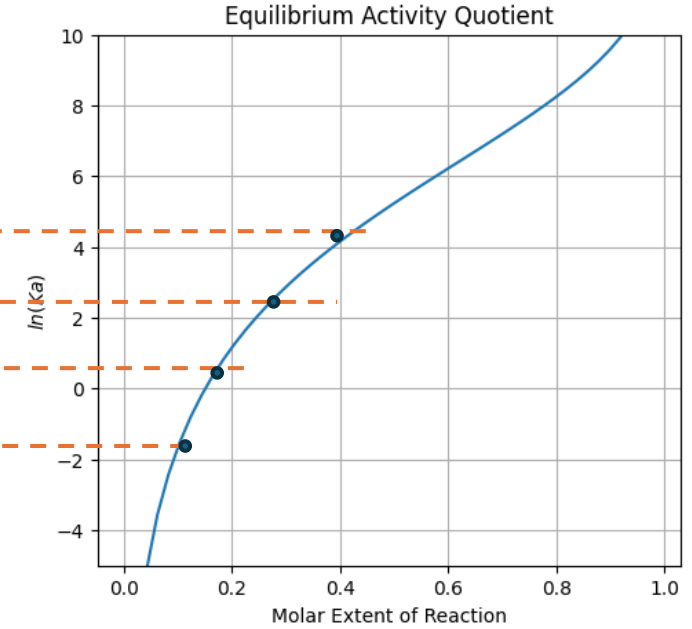
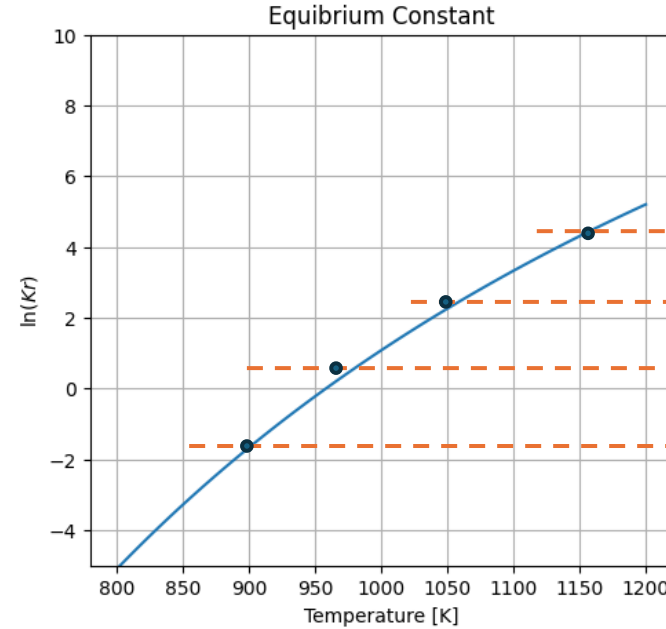
$$\ln K_r(T) = -\frac{1}{R} [\Delta G_{\text{rxn}}^\circ - \Delta H_{\text{rxn},298}^\circ + \Delta H_{\text{rxn},T}^\circ] \quad \text{.. (i)}$$

$$\ln K_a(\xi) = \sum_{c=1}^C \nu_c \ln (y_c(\xi) P) \quad \text{.. (ii)}$$

Equilibrium exists when: $\ln K_r(T) = \ln K_a(\xi)$

Variables

- Temperature
- Pressure
- $\text{H}_2\text{O}/\text{CH}_4$ ratio



Python model : *Steam_Metane_Reforming.ipynb*

Remarks & Improvements

- 93% of Methane conversion is achieved at $P = 1$ atm and H_2O/CH_4 ratio of 3.5
- $\Delta H_{1000} = 227 \text{ kJ mol}^{-1}$

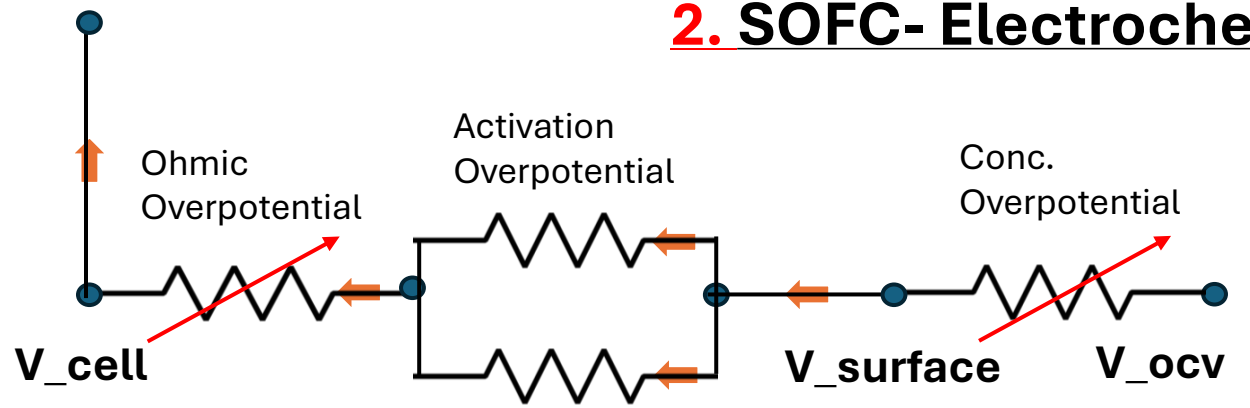
For flow rate of $CH_4 = 1 \text{ mol/s}$;

$$Q = 0.93 * 227 = 211 \text{ kJ s}^{-1} \text{ (Energy Requirement)}$$

This also helps to maintain H_2O concentration to prevent carbon deposition .

- Need to consider role of catalyst to aid Steam Methane Reforming rate.
- At high steam we can use pseudo 1st order reaction.
- Sizing can be done based on:
 - i. Power Requirement (Fuel flow rate)
 - ii. Kinetics of the reaction

2. SOFC- Electrochemical Resistances



Concentration Overpotential

$$V_{\text{conc,an},i} = \frac{RT_i}{nF} \left[\ln \left(1 + \frac{j_i}{i_{\text{conc,an},H_2O,i}} \right) + \ln \left(1 - \frac{j_i}{i_{\text{conc,an},i}} \right) \right]$$

$$V_{\text{conc,ca},i} = -\frac{RT_i}{2nF} \ln \left(1 - \frac{j_i}{2i_{\text{conc,ca},i}} \right)$$

$$i_{\text{conc,an},i} = \frac{nF(p_{H_2,i} \cdot 1000)D_{\text{anode}}}{RT_i L_{\text{anode}}}$$

$$i_{\text{conc,ca},i} = \frac{nF(p_{O_2,i} \cdot 1000)D_{\text{cathode}}}{RT_i L_{\text{cathode}}}$$

- If conc. overpotential is not ignored ; PDE's Boundary condition is driven by Electrochemical & Energy equations

Finite Difference Method

$$u_j \frac{C_j^{i+1} - C_j^i}{\Delta x} = D \frac{C_{j+1}^{i+1} - 2C_j^{i+1} + C_{j-1}^{i+1}}{\Delta y^2}$$

$$-\frac{D}{\Delta y^2} C_{j-1}^{i+1} + \left(\frac{u_j}{\Delta x} + \frac{2D}{\Delta y^2} \right) C_j^{i+1} - \frac{D}{\Delta y^2} C_{j+1}^{i+1} = \frac{u_j}{\Delta x} C_j^i$$

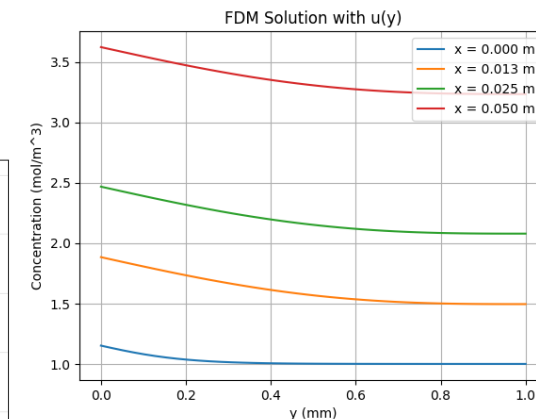
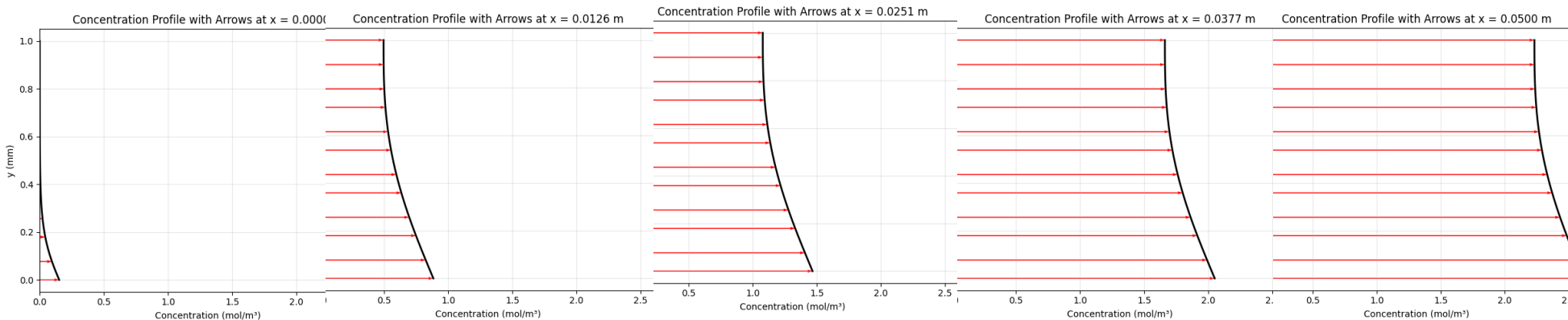
$$u \frac{\partial C}{\partial x} = D \frac{\partial^2 C}{\partial y^2}$$

Initial Condition: $C(0, y) = 0$

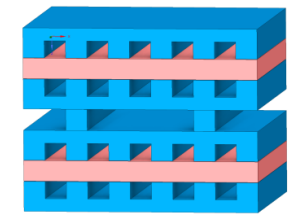
Boundary Condition 1: $-D \frac{\partial C}{\partial y} \big|_{y=0} = N_w$

or $j = 3000$;

Boundary Condition - 2: $D \frac{\partial C}{\partial y} \big|_{y=1 \text{ mm}} = 0$;



Improvement in Work output with Multi-staging



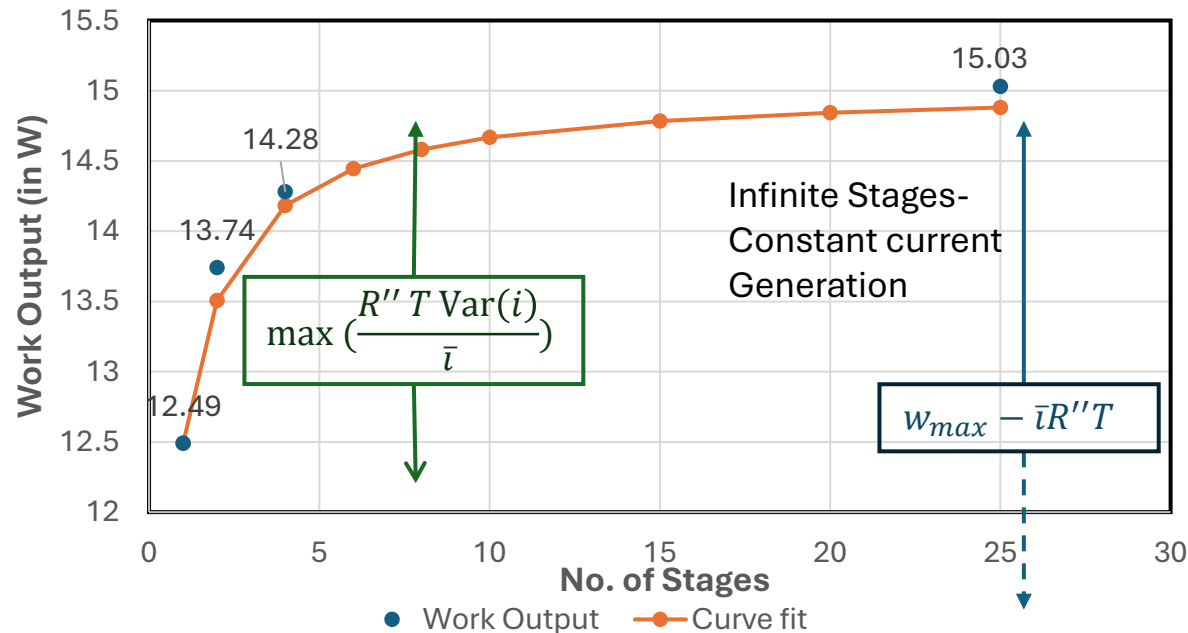
Improving the Total Work output per Unit Consumption
(Without any Heat Transfer)

Subject to constraints:

- Consumption of fuel = 63 %
- Area of the fuel cell = 80 cm²
- Temperature at each cell > 837 K

$$W_{\text{output}} = w_{\text{max}} - \bar{i}R''T - \frac{R''T \text{Var}(i)}{\bar{i}}$$

Equipartition



Heat Engine Efficiency:

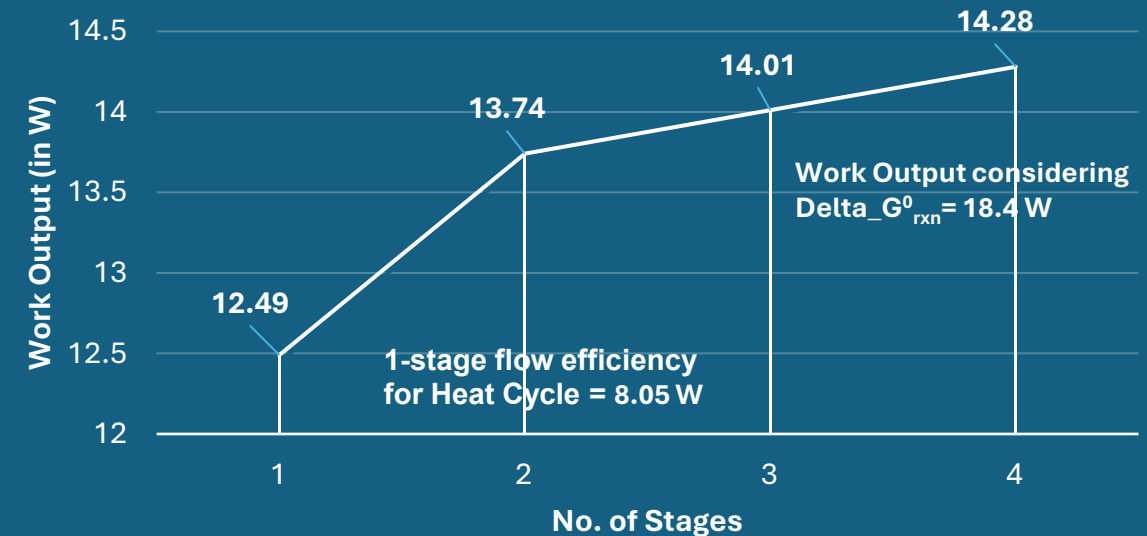
$$r = T_L/T_H ; T_L = 303 \text{ K}; T_H = 1100 \text{ K}$$

$$\eta = \left(1 - \sqrt{T_L/T_H}\right) / \left(1 + \sqrt{T_L/T_H}\right) = 31.3\%$$

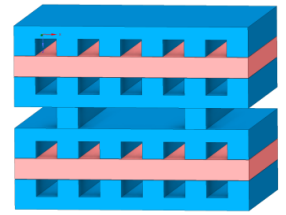
Enthalpy of combustion = -143 kJ/g

Work Output= 8.05 J/s

WORK PER UNIT CONSUMPTION WITH EQUI-PARTITION



Multi-staging of V_{cell}: Without Heat Transfer



Work output (Electrical + Sensible Heat):

Assuming Single stage efficiency of heated output

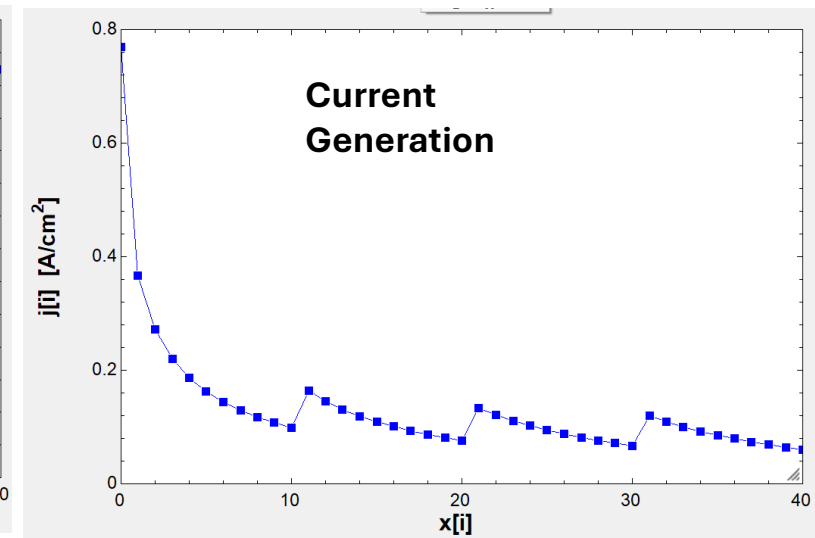
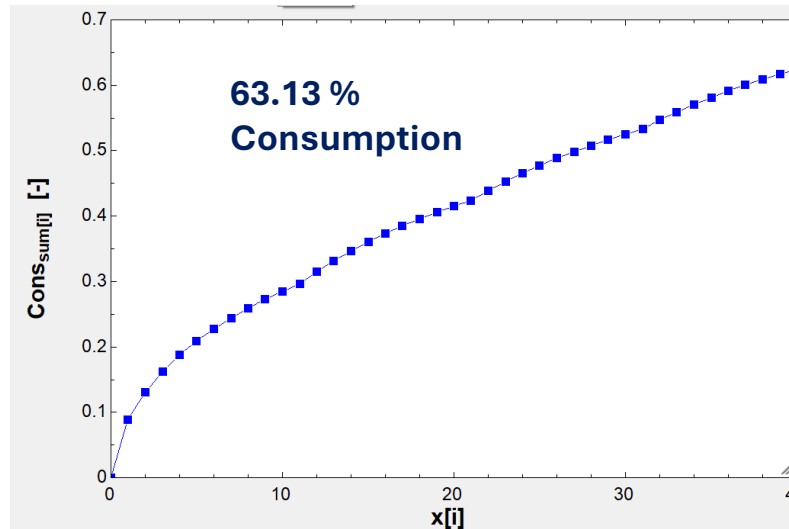
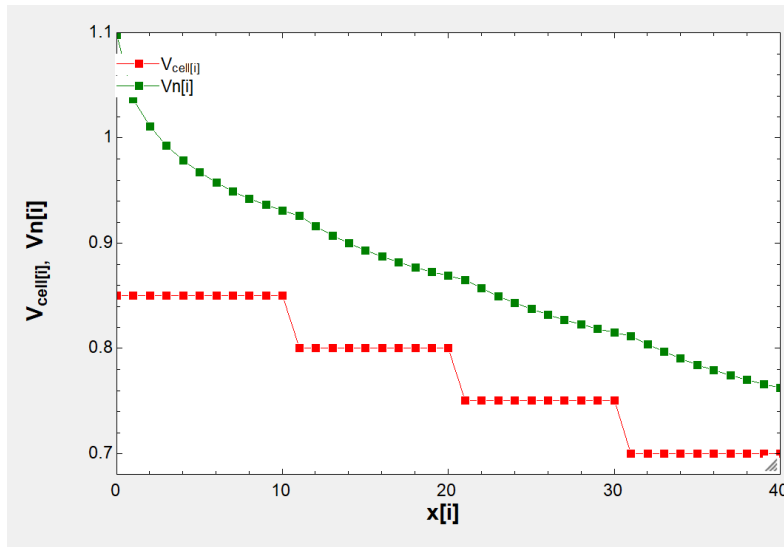
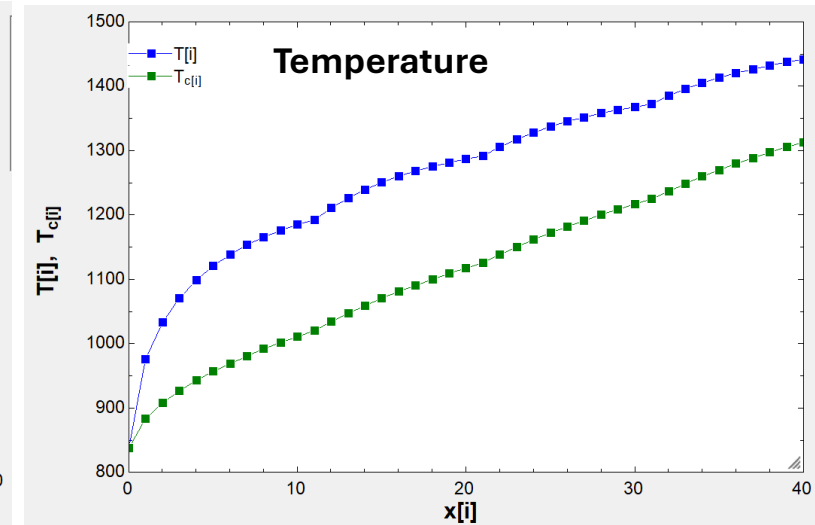
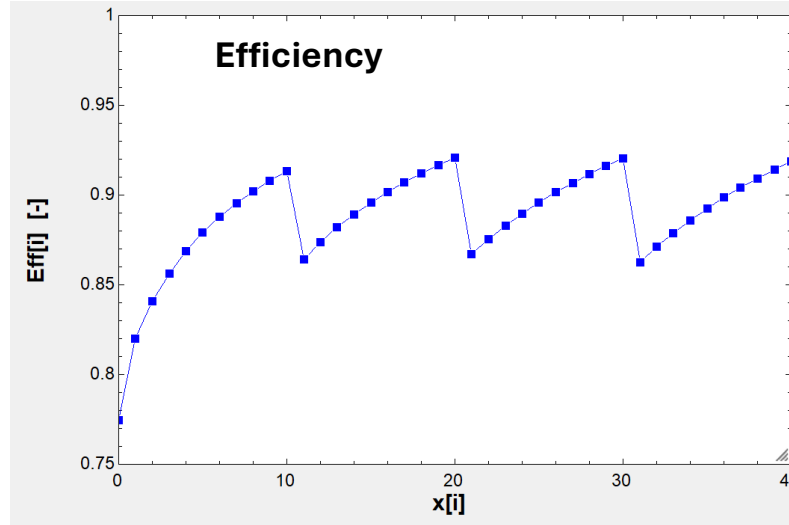
$$\frac{\text{Work}}{\text{Fuel Consumption}} = \frac{14.28 + 1.128 \text{ J/s}}{0.00018 \text{ g/s}} = 85.6 \text{ kJ/g}$$

Enthalpy of combustion = -143 kJ/g

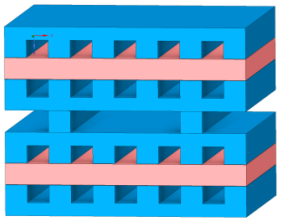
Delta_G⁰_{rxn} = -102.6 kJ/g

Efficiency = 59.86 %: For 63.13 % Consumption

Consumption Weighted Efficiency =
(0.63 * 59.86) + (0.368 * 31.3) = 49.25%



Multi-staging of V_cell: Without Heat Transfer- Higher Consumption



Work output (Electrical + Sensible Heat):

Assuming Single stage efficiency of heated output

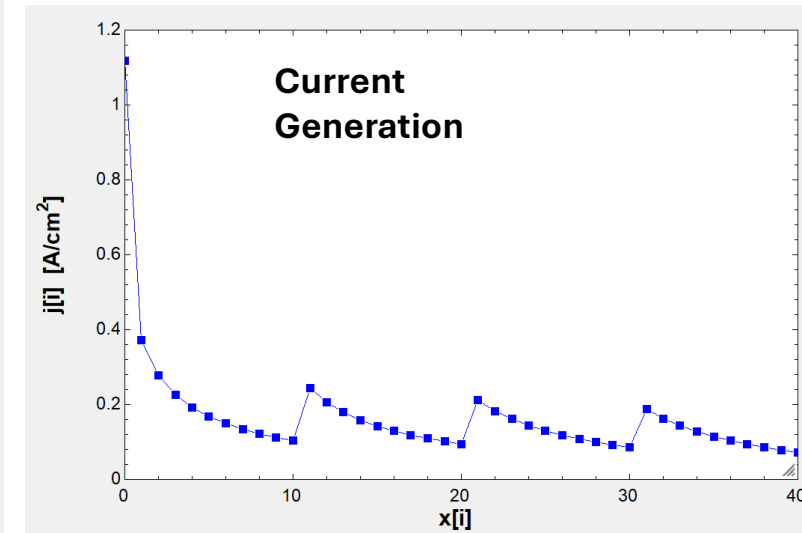
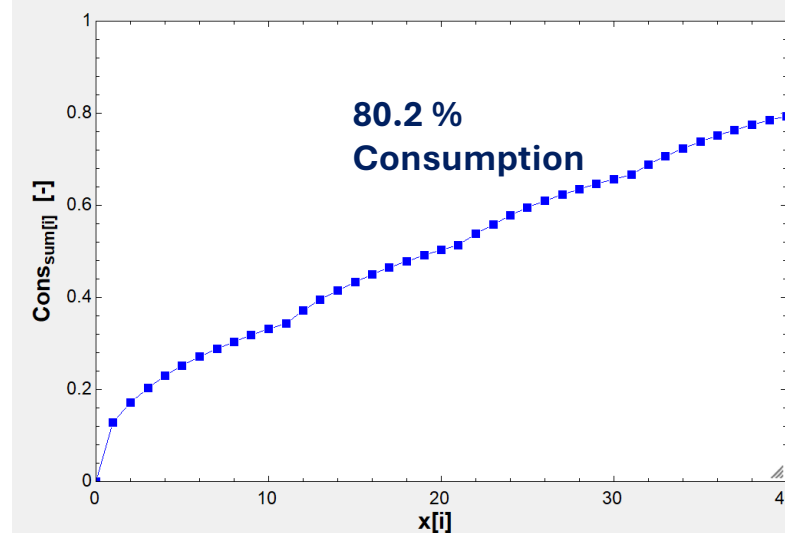
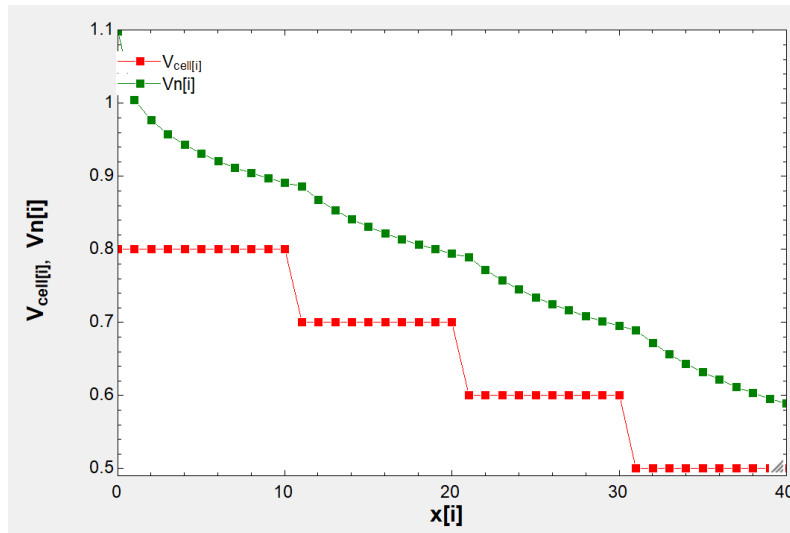
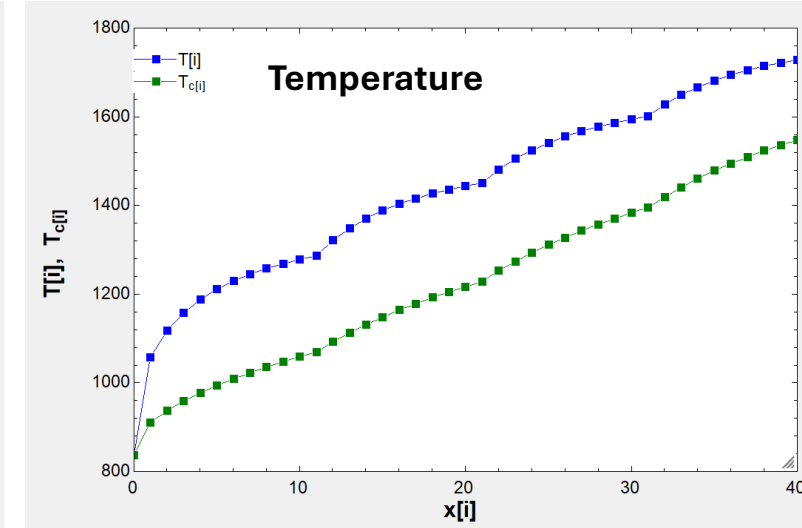
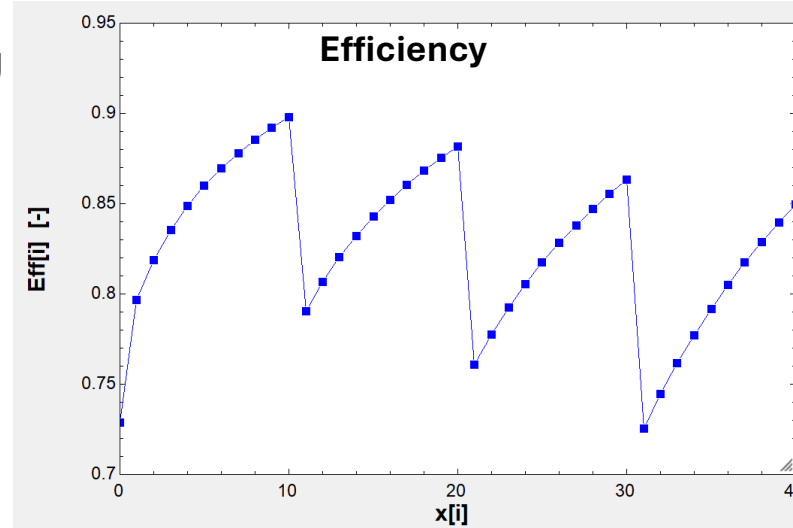
$$\frac{\text{Work}}{\text{Fuel Consumption}} = \frac{12.75 + 1.84 \text{ J/s}}{0.00018 \text{ g/s}} = 81.055 \text{ kJ/g}$$

Enthalpy of combustion = -143 kJ/g

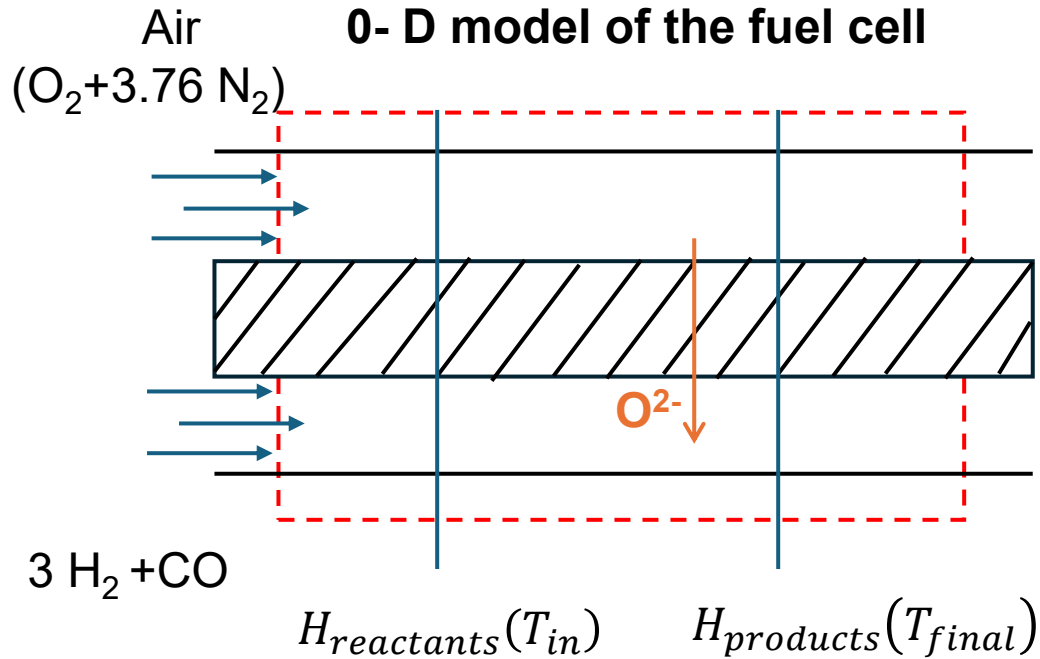
Delta_G⁰_{rxn} = -102.6 kJ/g

Efficiency = 56.68 % : For 80.2 % Consumption

Consumption Weighted Efficiency =
(0.802 * 56.68) + (0.198 * 31.3) = 51.65 %



Mathematical model for Fuel Cell – CO+ H₂ as Fuel



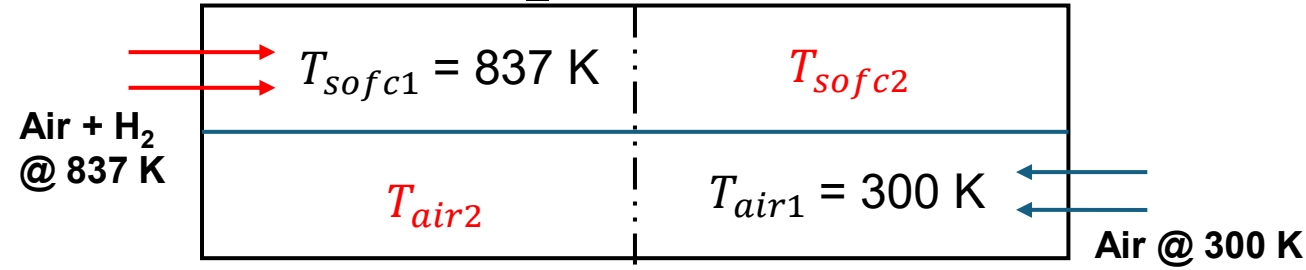
Activation Overpotential

$$j_{CO} = j_{0,CO} \left[\exp\left(\frac{\alpha_a n F \eta_{CO}}{RT}\right) - \exp\left(-\frac{\alpha_c n F \eta_{CO}}{RT}\right) \right] \dots (ix)$$

$$j_{H_2} = j_{0,H_2} \left[\exp\left(\frac{\alpha_a n F \eta_{H_2}}{RT}\right) - \exp\left(-\frac{\alpha_c n F \eta_{H_2}}{RT}\right) \right] \dots (x)$$

$$j^i = j_{0,ca} \left[\exp\left(\frac{0.5 n F \eta_{act,ca}}{RT_i}\right) - \exp\left(-\frac{0.5 n F \eta_{act,ca}}{RT_i}\right) \right] \dots (xi)$$

$$V_{ocv} - V_{cell} = \eta_{act,ca} + \eta_{act,an} \dots (xii)$$



Energy Balance:

$$H_{reactants}(T_{in}) - H_{products}(T_{final}) = W_e^i \dots (i)$$

$$\eta^i = \frac{V_{cell}}{V_{ocv}(T)} \dots (ii)$$

$$W_e^i = -\frac{\eta^A}{nF} (j_{H_2} \Delta G_{H_2} + j_{CO} \Delta G_{CO}) \dots (iii)$$

$$V_{H_2}(T) = \frac{\Delta G_{H_2}(i)}{nF} + \frac{RT_i}{nF} \ln \left(\frac{\left(\frac{p_{H_2,i}}{P}\right) \left(\frac{p_{O_2,i}}{P}\right)^{1/2}}{\left(\frac{p_{H_2O,i}}{P}\right)} \right) \dots (iv)$$

$$V_{CO}(T) = \frac{\Delta G_{CO}(i)}{nF} + \frac{RT_i}{nF} \ln \left(\frac{\left(\frac{p_{CO,i}}{P}\right) \left(\frac{p_{O_2,i}}{P}\right)^{1/2}}{\left(\frac{p_{CO_2,i}}{P}\right)} \right) \dots (v)$$

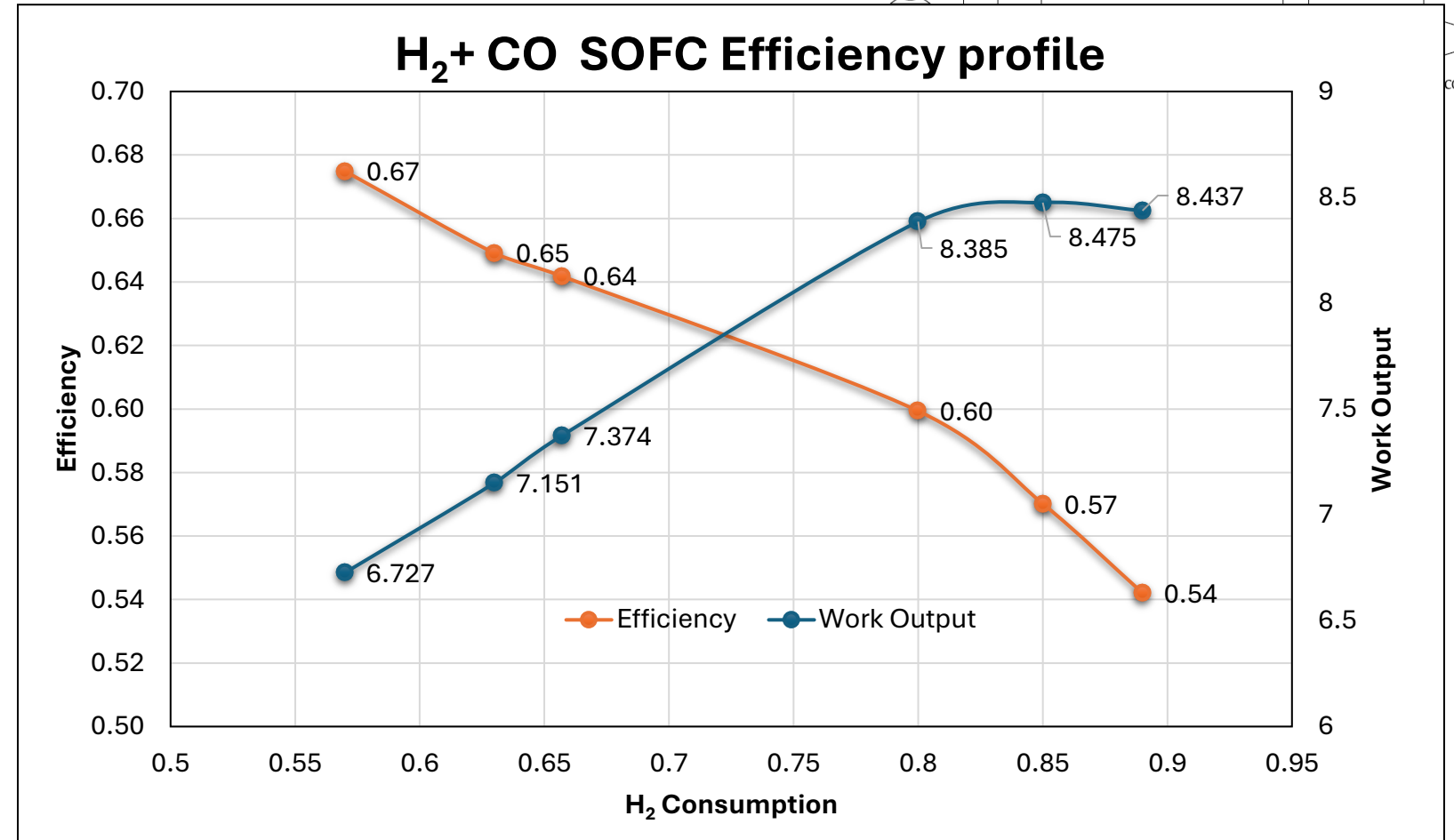
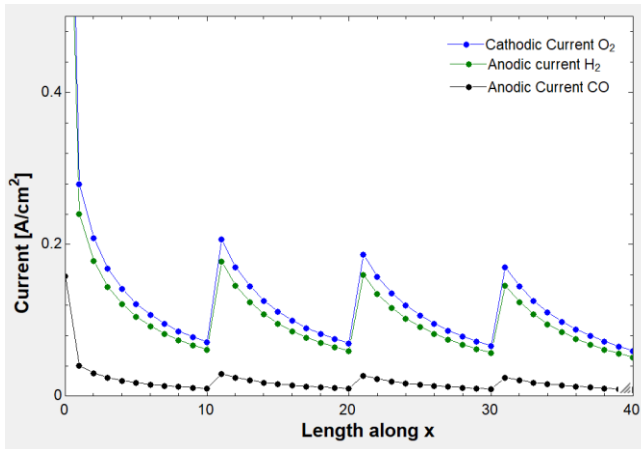
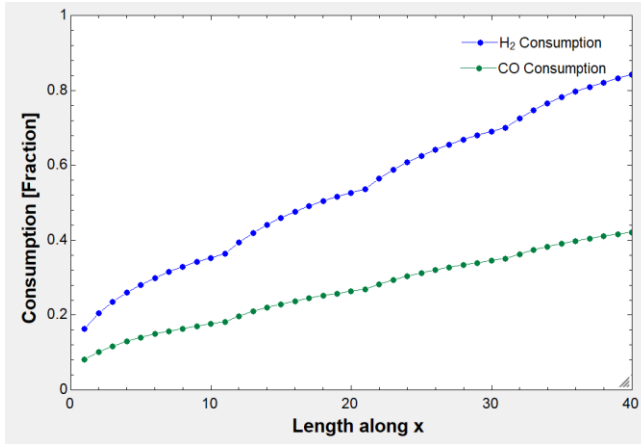
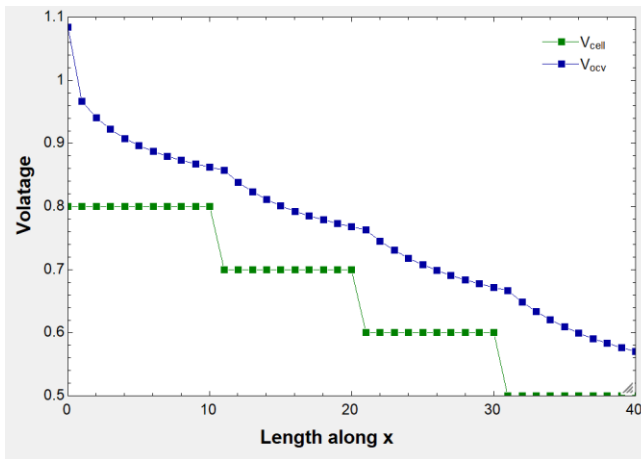
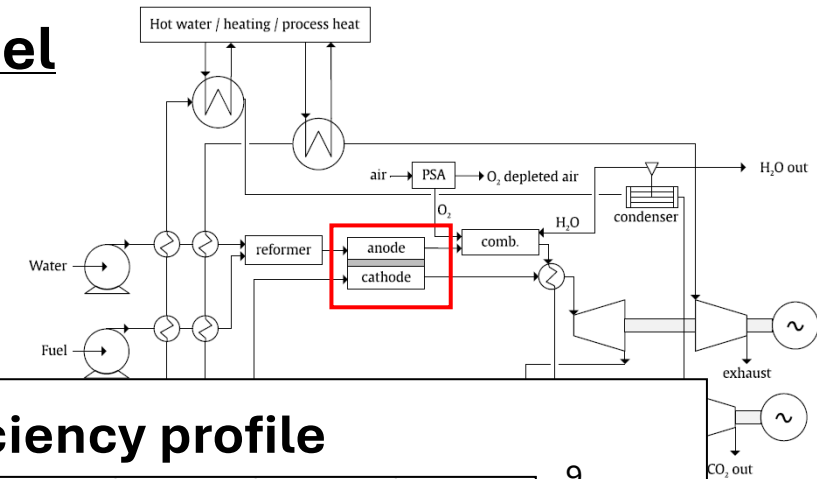
$$j = j_{H_2} + j_{CO} \dots (vi) \quad \eta_{CO} = \eta_{H_2} = \eta_{an} \dots (vii)$$

$$V_{ocv}(T) = \frac{j_{H_2} V_{H_2} + j_{CO} V_{CO}}{j_{H_2} + j_{CO}} \dots (viii)$$

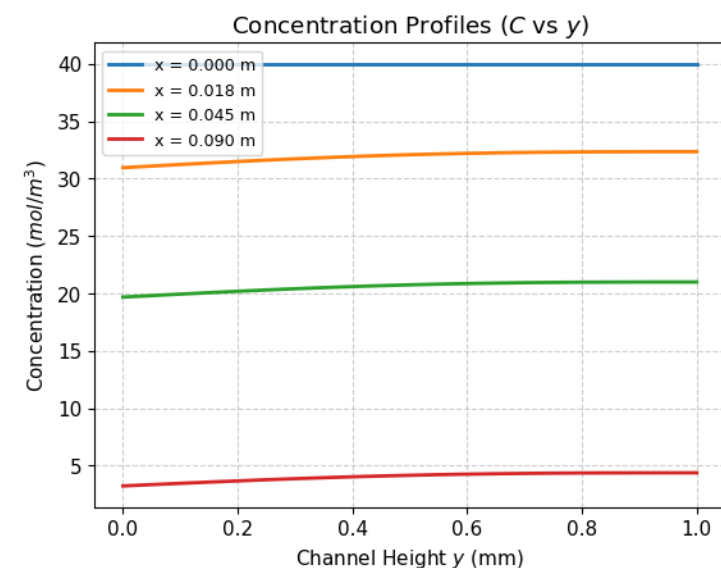
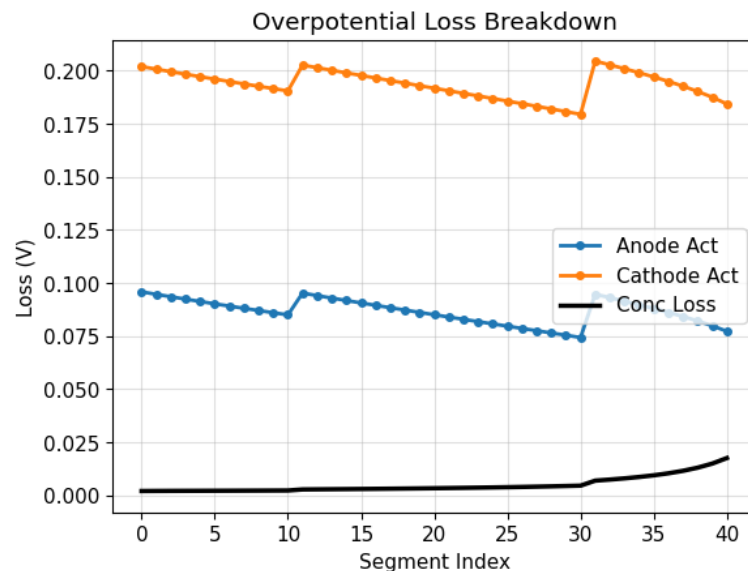
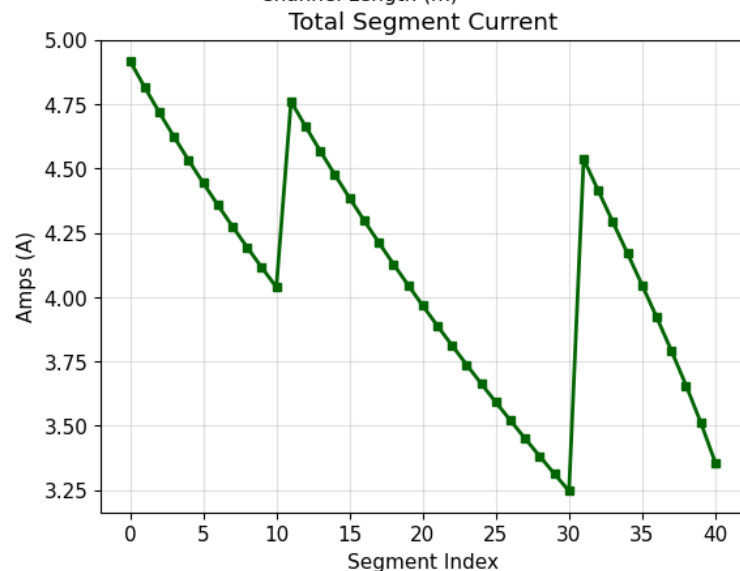
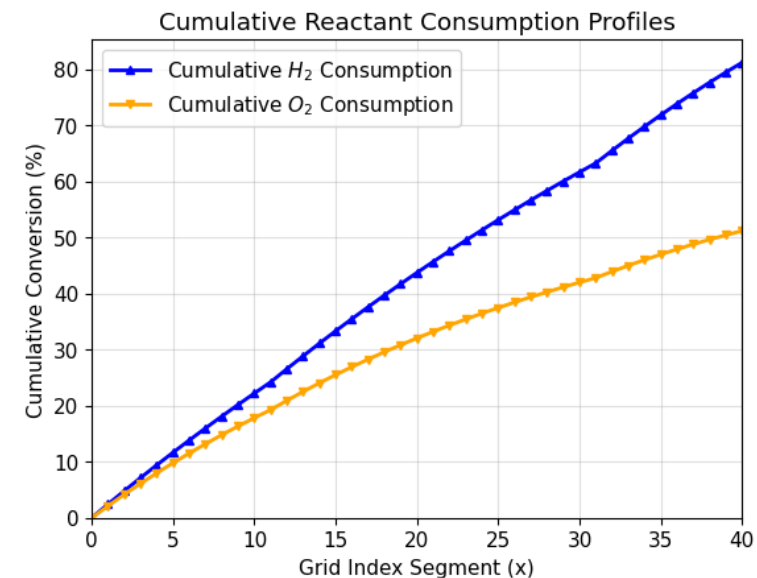
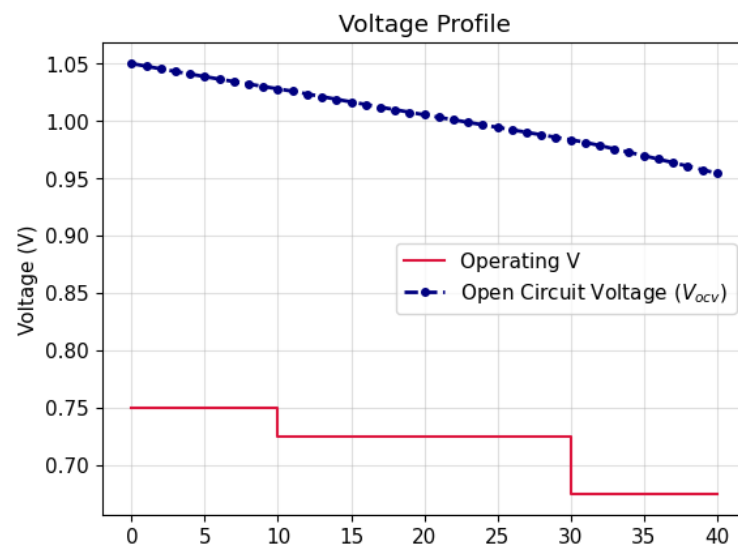
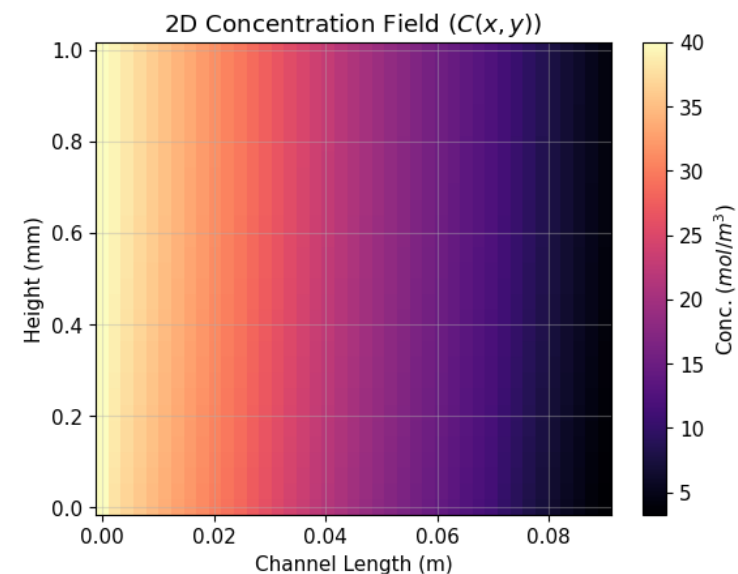
Solid Oxide Fuel Cell – CO+ H₂ Fuel

Variables

- Effect of fractional consumption
- Effect of Heat Transfer



PDE constrained Solutions– CO+ H₂ Fuel



Remarks & Improvements

Work output (Electrical):

$$\frac{\text{Work}}{\text{Fuel Consumption}} = \frac{8.458 \text{ J/s}}{(0.425 \times 0.00002 \times 28) + (0.85 \times 0.00006 \times 2) \text{ g/s}} = \frac{8.458 \text{ J/s}}{0.00034 \text{ g/s}} = \mathbf{24.876 \text{ kJ/g}}$$

Enthalpy of combustion = **-43.23 kJ/g**

$$\begin{aligned} \Delta G_{\text{rxn}}^0 &= -943 \text{ kJ/mol} \\ &= \mathbf{-40.6 \text{ kJ/g} \text{ ! Ratio of fuel consumption is different}} \end{aligned}$$

Efficiency = 57.5 %

$$\frac{\text{Work}}{\text{Mole of methane}} = \frac{8.458 \text{ J/s}}{0.00002 \text{ mol/s}} = \mathbf{422.9 \text{ kJ/mol}}$$

$$\dot{W} = \mathbf{422.9 \text{ kJ s}^{-1} \text{ (Work Output)}}$$

- Assuming incomplete conversion of Methane, we need to account for Internal Reforming in the fuel cells.
- Since H_2 and CO has different i_{an}^0 , is there away to tweak it to get equal fractional conversion over time.
- Is there any contribution due to tortuosity or the porosity of the fuel cell electrodes? Does it play a role in determining the Diffusion coefficient in the model?
- The variables of the Fuel cell can be listed as
 - i. Mass flow rate in the duct of SOFC
 - ii. V_{cell} of the SOFC (It eventually drives the current per unit area.
 - iii. Length of the Fuel cell (It can decide the fractional consumption of fuel.

Also, a larger area would need larger cell increasing the Capex of the project.

3. Recompression- Reheat Cycle

Turbine & Compressor

$$h_i = h(F^\$, T = T_i, P = P_H) \text{ for } i = 1 \text{ to } 12$$

$$s_i = s(F^\$, T = T_i, P = P_H) \text{ for } i = 1 \text{ to } 12$$

$$s_2 = s_1$$

$$s_6 = s_5$$

$$s_3 = s_8$$

High Temp. Recuperator & Low Temp. Recuperator

$$\dot{m}_6 \cdot (h_6 - h_7) = \dot{m}_4 \cdot (h_4 - h_{12})$$

$$\dot{Q}_{HTR} = \dot{m}_6 \cdot (h_6 - h_7)$$

$$Avg_{TD,HTR} = \frac{(T_6 - T_4) + (T_7 - T_{12})}{2}$$

$$U_{HTR} \cdot A_{HTR} \cdot Avg_{TD,HTR} = \dot{Q}_{HTR}$$

$$\dot{m}_2 \cdot (h_{11} - h_2) = \dot{m}_4 \cdot (h_7 - h_8)$$

$$\dot{Q}_{LTR} = \dot{m}_4 \cdot (h_7 - h_8)$$

$$Avg_{TD,LTR} = \frac{(T_7 - T_{11}) + (T_8 - T_2)}{2}$$

$$U_{LTR} \cdot A_{LTR} \cdot Avg_{TD,LTR} = \dot{Q}_{LTR}$$

$$\dot{m}_{11} \cdot h_{11} + \dot{m}_3 \cdot h_3 = \dot{m}_{12} \cdot h_{12}$$

$$\dot{S}_{gen,mixing} = \dot{m}_{12} \cdot s_{12} - (\dot{m}_{11} \cdot s_{11} + \dot{m}_3 \cdot s_3)$$

Mixing at 3

$$\eta = \dot{W}_{net} / \dot{Q}_{in}$$

$$\dot{S}_{gen,total} = \dot{S}_{gen,LTR} + \dot{S}_{gen,HTR} + \dot{S}_{gen,heater} + \dot{S}_{gen,cooler} + \dot{S}_{gen,mixing}$$

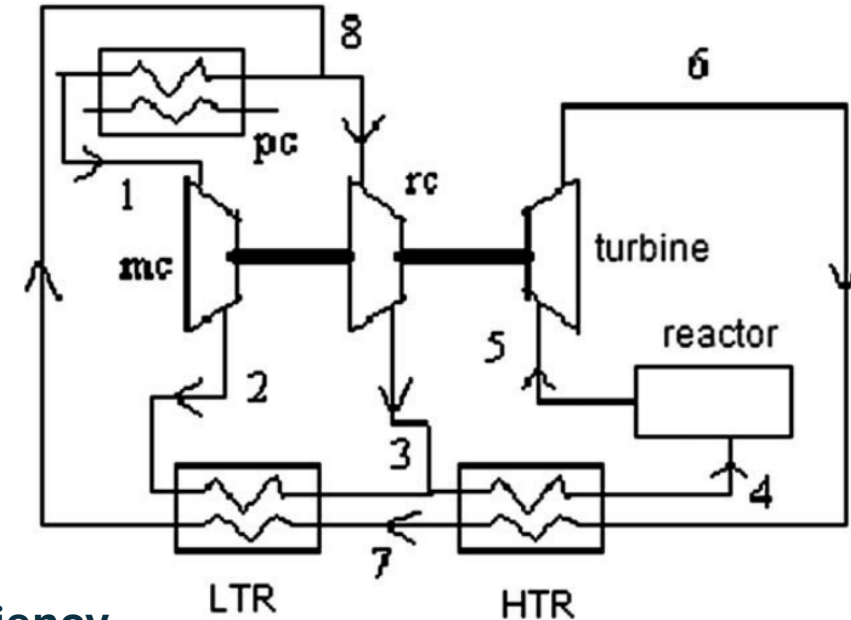
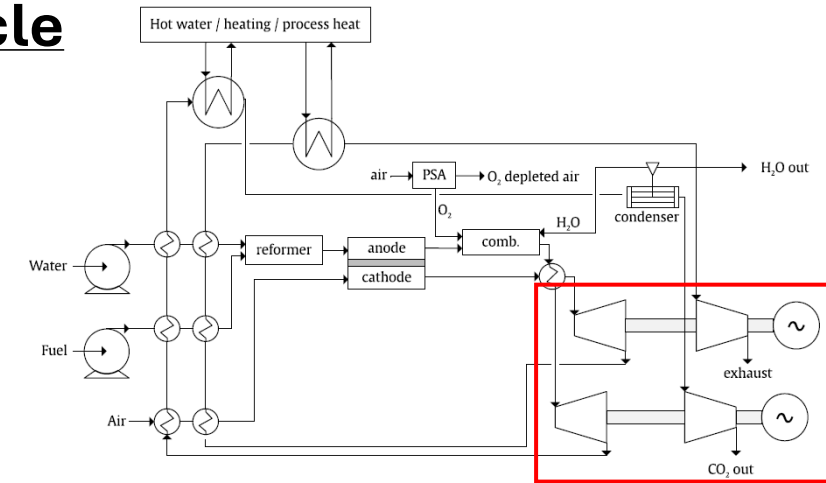
Update:

- Efficiency

Update:

- Discretization
- LMTD (Log Difference)

Efficiency and Entropy generation

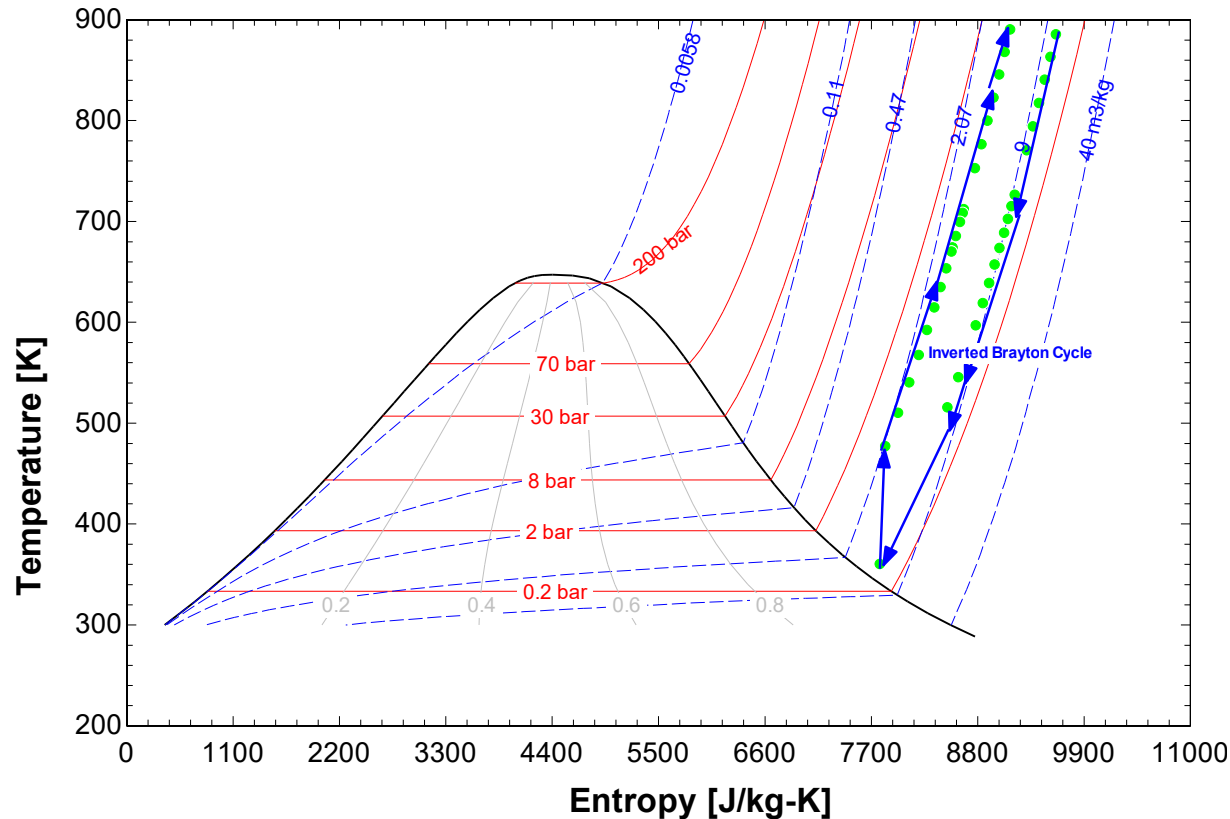


Inverted Brayton Cycle –Thermo Model

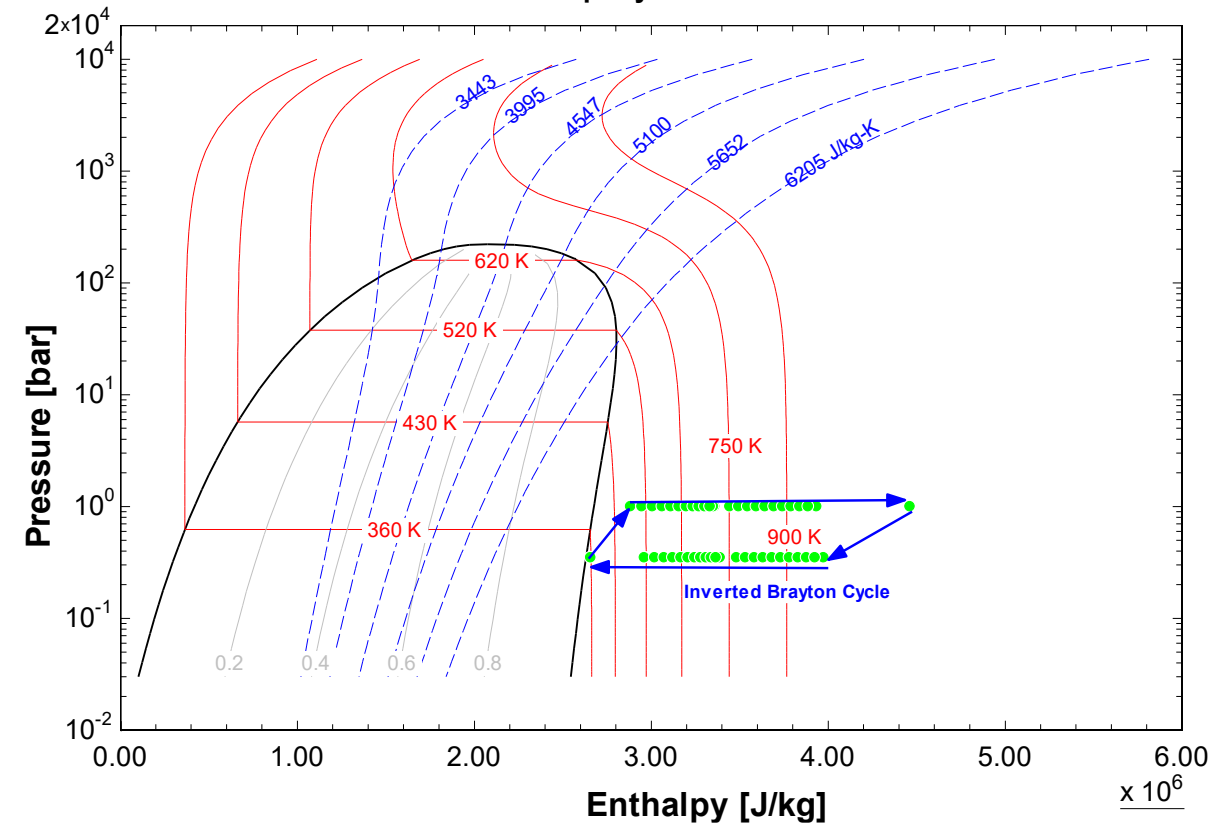
- Pressure drop and heat transfer coefficient–
Modelled using "**Turbulent Flow - Gnielinski equation**"
Diameter= 3.3 cm
- Area of HTR = 0.4 m²
Area of LTR = 0.2 m²
- Isentropic efficiency:
Turbine = 93 % ;
Compressor = 88 %

- frac_recomp = 0.1
- (m_{rc}/m_{tot})
- $T_{hot} = 1200\text{ K}$
- $T_{cold} = 360\text{ K}$
- $P_{high} = 1\text{ bar}$
- $P_{low} = 0.35\text{ bar}$
- Discrete points in HX= 10

T-s Property Diagram - Steam



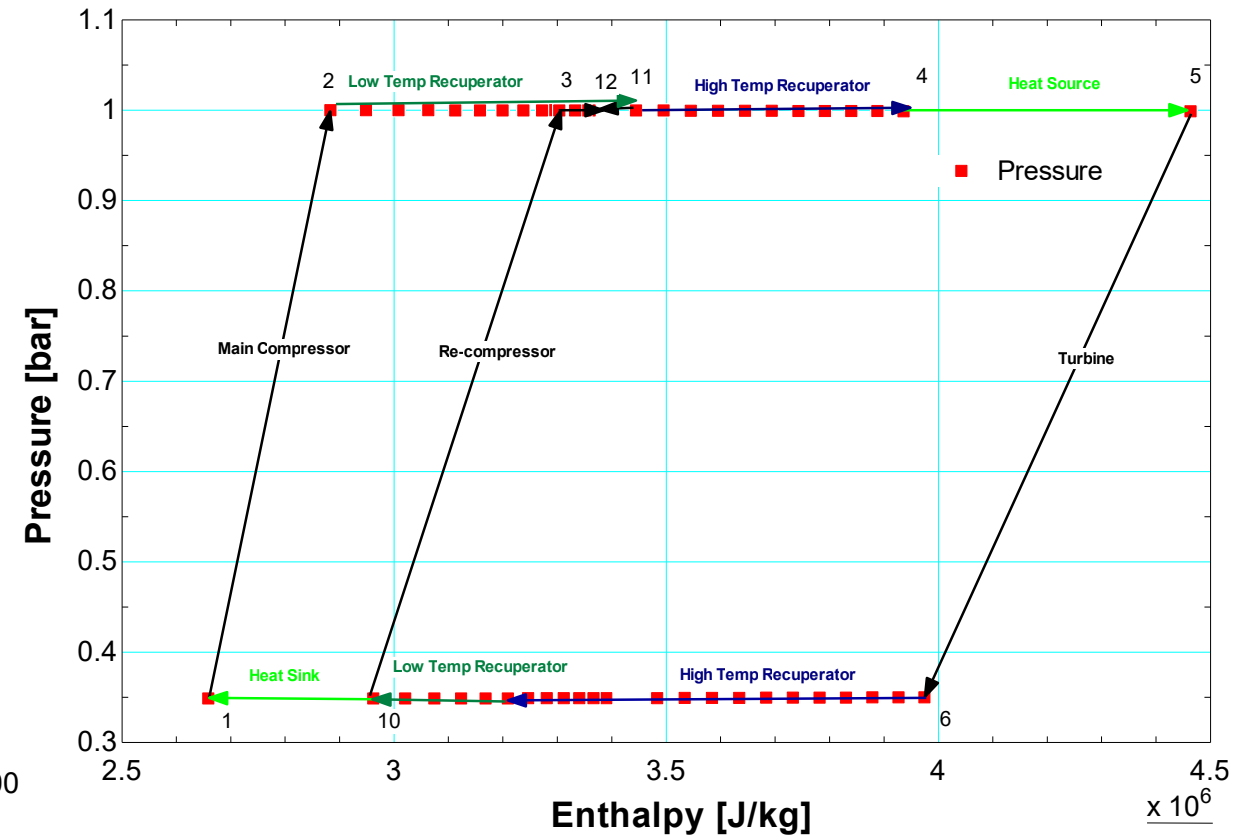
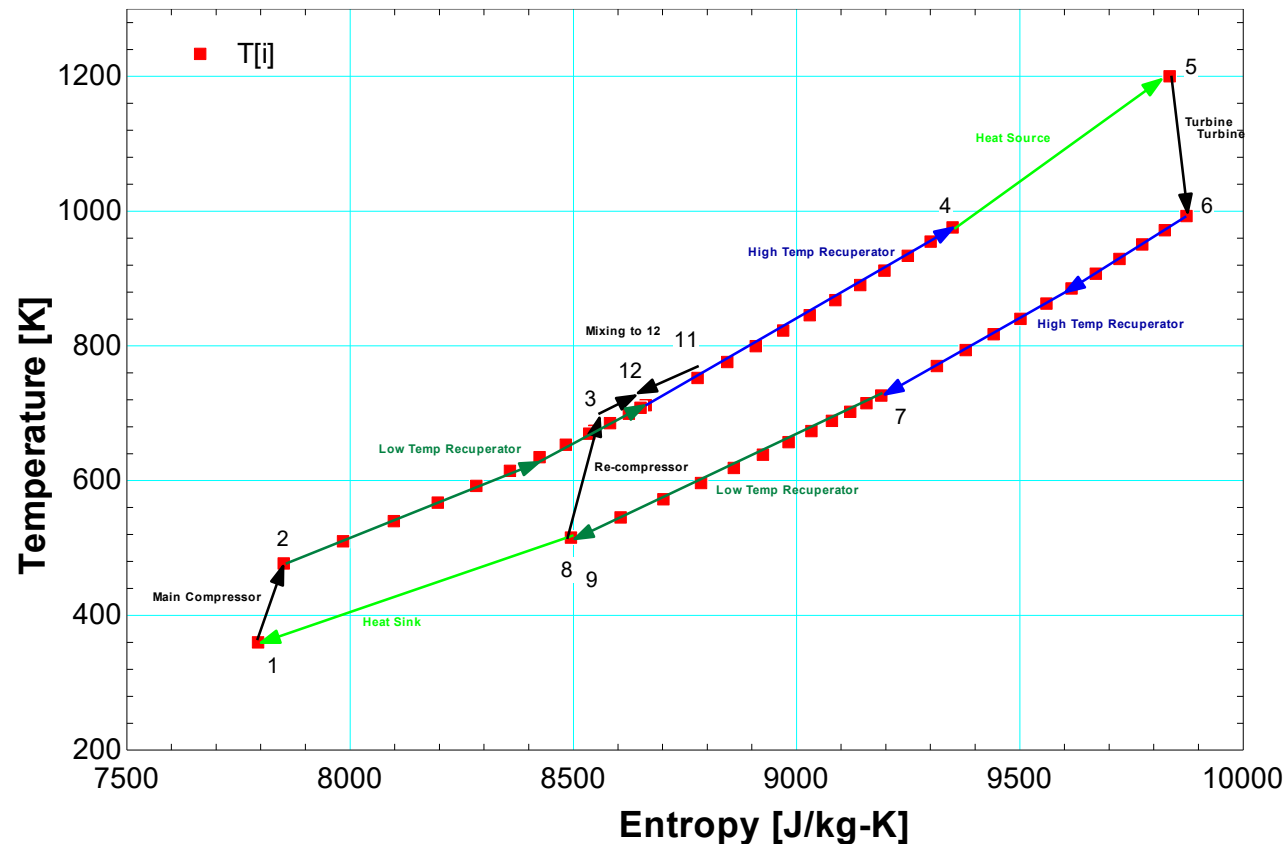
P-h Property Plot - Steam



Inverted Brayton Cycle –Thermo Model

Results

- Efficiency = 48.3 %
- $P_{\text{drop1}} = 0.0047 \text{ bar}$
- $P_{\text{drop2}} = 0.0021 \text{ bar}$
- $P_{\text{drop3}} = 0.0023 \text{ bar}$
- $U_{\text{HTR}} (\text{avg}) = 779 \text{ W/m}^2 \text{ K}$
- $U_{\text{LTR}} (\text{avg}) = 880 \text{ W/m}^2 \text{ K}$
- $W_{\text{turb}} = 4886 \text{ W}$
- $W_{\text{comp1}} = 2018 \text{ W}$
- $W_{\text{comp2}} = 318 \text{ W}$
- $S_{\text{gen cooler}} = 1.263 \text{ W/K}$
- $S_{\text{gen heater}} = 0.46 \text{ W/K}$
- $S_{\text{gen HTR}} = 0.16 \text{ W/K}$
- $S_{\text{gen LTR}} = 0.34 \text{ W/K}$
- $S_{\text{gen mixing}} = 0.0022 \text{ W/K}$



Remarks & Improvements

Work output (Sensible Heat):

Enthalpy of combustion = **-15.96 kJ/g**

Efficiency = 48.3 %

$$\frac{\text{Work}}{\text{Mole of methane}} = \frac{5.43 * 0.483 \text{ J/s}}{0.00002 \text{ mol/s}} = \frac{2.622 \text{ J/s}}{0.00002 \text{ mol/s}} = \mathbf{131.1 \text{ kJ/mol}}$$

$$\dot{W} = \mathbf{131.3 \text{ kJ s}^{-1} \text{ (Work Output)}}$$

- Need to check if the material will be able to withstand higher temperatures.
- Increasing the heat output rate can decrease the efficiency further. We need to pump heat out at higher delta T to get higher Work output
- Is it possible to use H2O-CO2 mixture as fluid for heat transfer. CO2 might give some advantages in terms of lower compressor work?
- Is there some limitations on sizing of the Heat Exchanger?

4. Work for CO₂ Separation

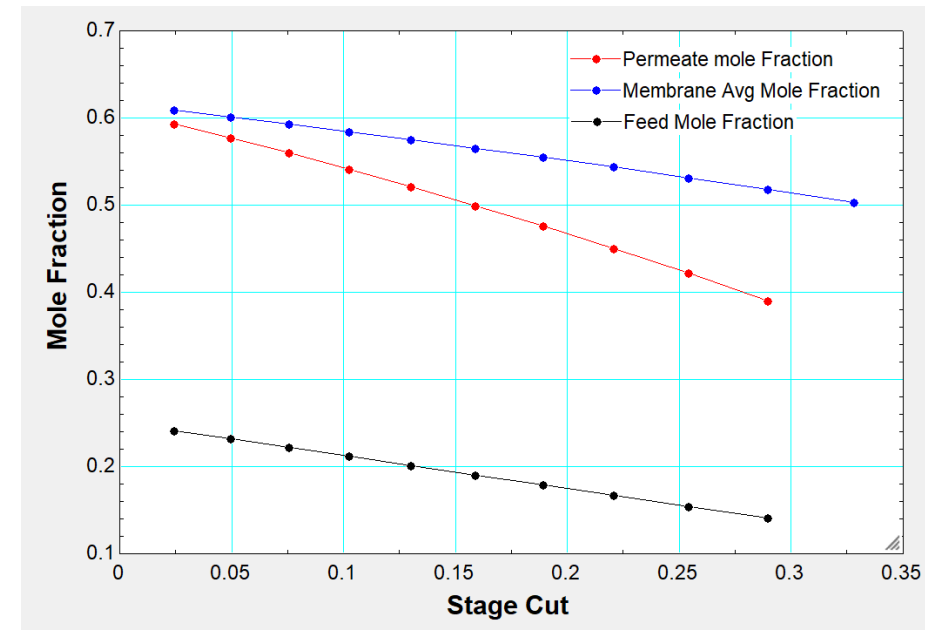
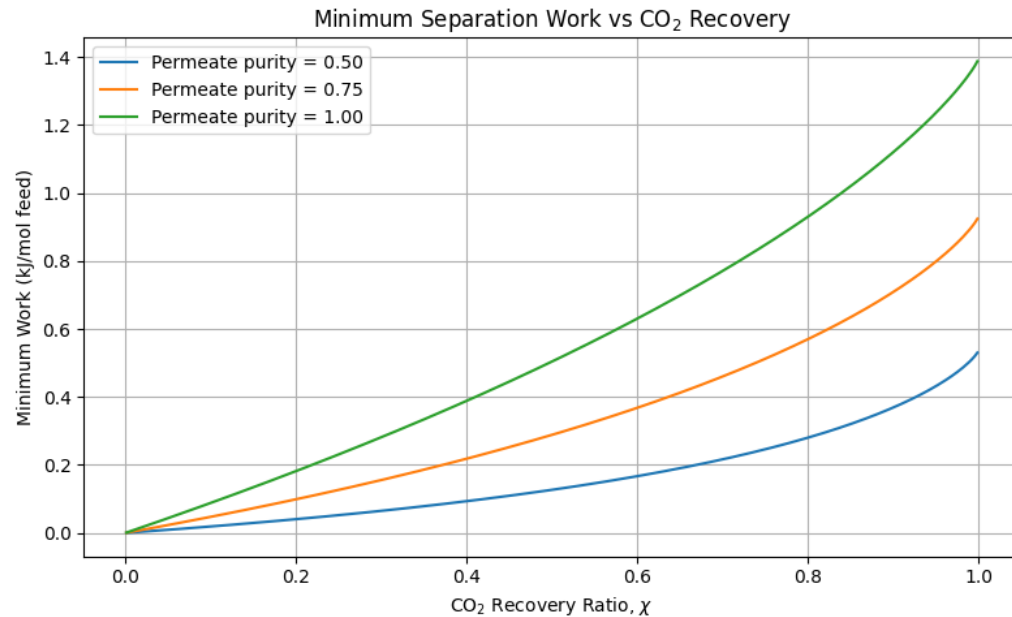
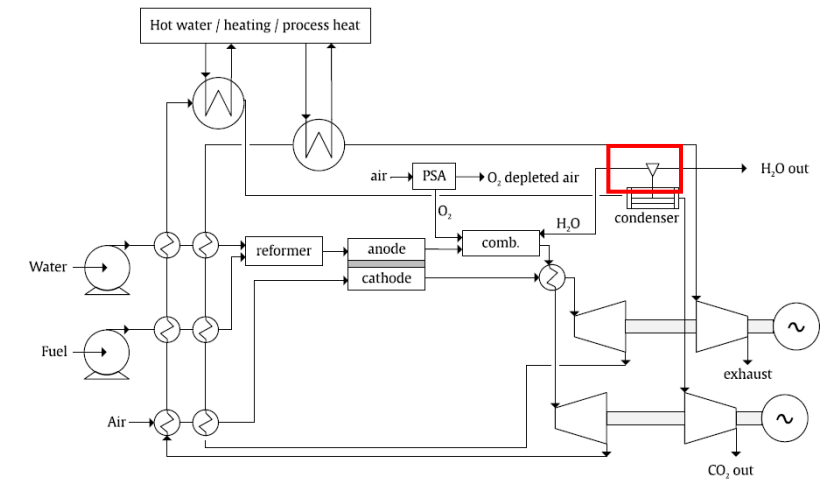
For dilute CO₂ Removal

$$\frac{W_{min}}{RT} = \theta[y \ln(y) + (1 - y) \ln(1 - y)] + (1 - \theta)[x \ln(x) + (1 - x) \ln(1 - x)] - [z \ln(z) + (1 - z) \ln(1 - z)] \quad ; \quad \theta = \frac{\chi z}{y}$$

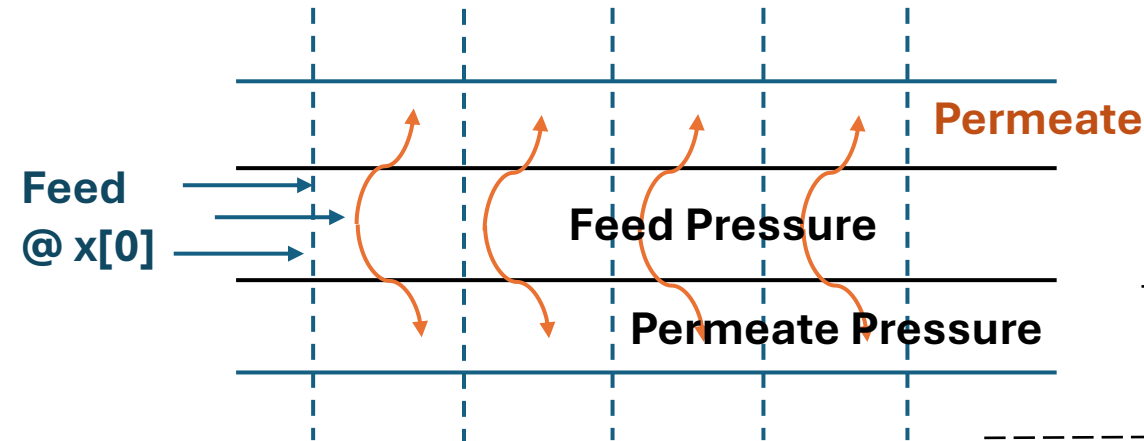
For 100% pure CO₂ Removal

$$\frac{W}{\Delta n_i RT} = -\chi \ln y_{iA} + (1 - \chi) \ln(1 - \chi) + \frac{1 - \chi y_{iA}}{y_{iA}} \ln \left(1 + \frac{\chi y_{iA}}{1 - \chi y_{iA}} \right)$$

$W_{\min}$ = Minimum reversible separation work y_{iA} = initial mole fraction in feed

$$\Delta n_i = \text{moles of separated component recovered} \quad \chi = \text{Recovery ratio}$$


CO₂ Separation :Dual Sorption Model



$$\chi_{i+1} = \frac{N_{CO_2,i+1}}{n_0 x_0}$$

$$\theta_{i+1} = \frac{V_{i+1}}{n_0}$$

$$(\alpha - 1)(y_i)^2 + \left(1 - \alpha - \frac{1}{r} - \frac{x(\alpha - 1)}{r}\right)y_i + \frac{\alpha x}{r} = 0 \quad c = c_H + c_L = Hp + \frac{c'_L bp}{1 + bp}$$

$$y_i = \frac{J_{CO_2,i}}{J_{CO_2,i} + J_{H_2O,i}}$$

$$\Delta \dot{V}_i = \frac{\dot{n}_i(x_i - x_{i+1})}{y_i - x_{i+1}}$$

$$J_{CO_2,i} = \Delta \dot{V}_i y_i$$

$$\Delta A_i = \frac{\Delta \dot{V}_i y_i}{J_{CO_2,i}}$$

c = total sorbed concentration

ΔA_i = Area of Membrane

$J_{i,i}$ = Unit Flow rate

H = Henry's constant

c'_L = Langmuir capacity constant

b = Langmuir affinity constant

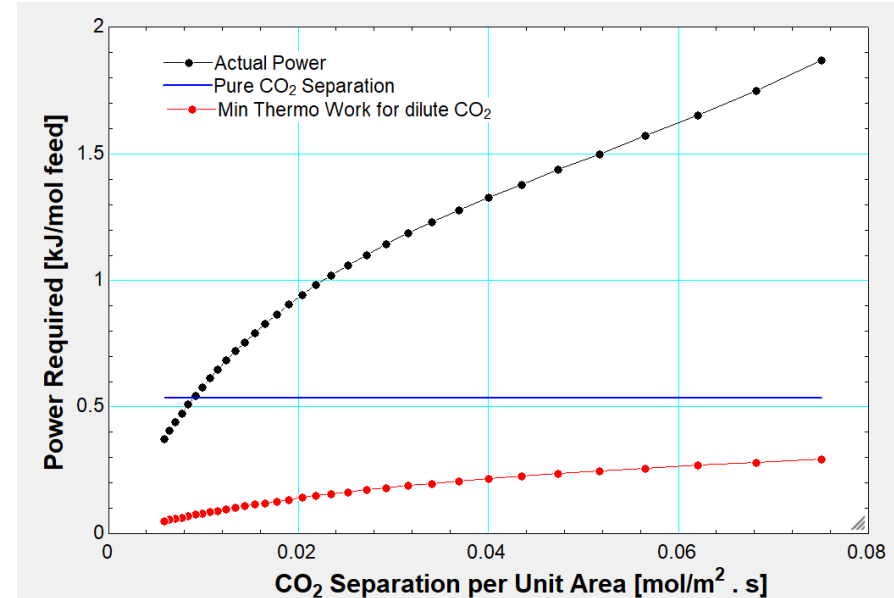
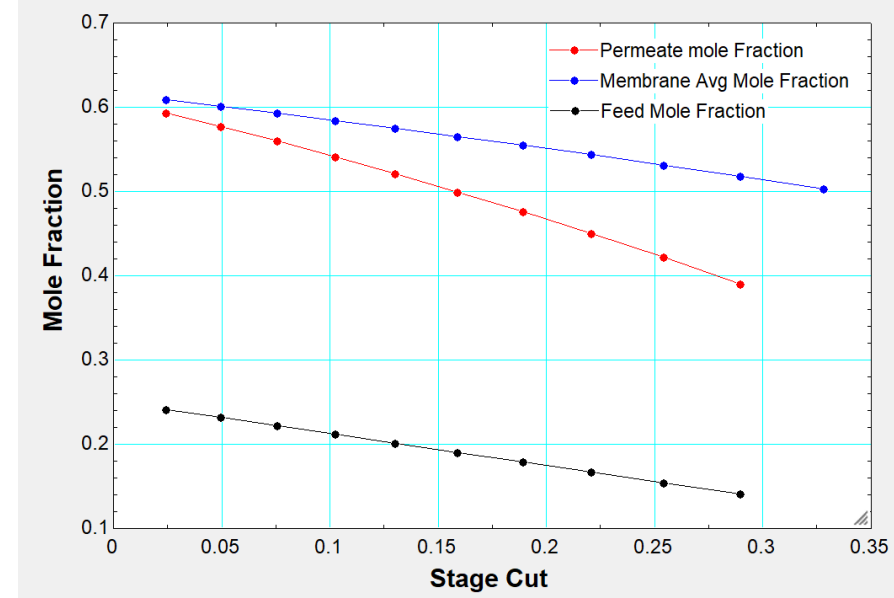
p = gas pressure

r = pressure ratio = $\frac{p_p}{p_f}$,

α = membrane selectivity,

x = feed mole fraction permeable gas,

y^* = permeate mole frac of perm. gas.



Efficiency of System

Considering flow rate of Methane as 1 mol/s

Energy Accounting	Heat of Combustion	Efficiency	Net Energy	
Work Output (Solid Oxide Fuel Cell)	43.23 kJ/g	57.50%	+422.9	kJ/s
Work Output (Inverted Brayton Cycle)	15.96 kJ/g	48.30%	+131.1	kJ/s
Heat Requirement (Steam Reforming)	227 kJ/mol	93 % conv.	-211.0	kJ/s
Energy Requirement (CO₂ separation)			-6.0	kJ/s
			337.0	kJ/s

Enthalpy of Combustion (Methane)= 802.3 kJ/mol

Efficiency = 337/ 802.3 = 42%